Lab 8: Linear Transformations, Inverses, and Projections

EECS 245, Fall 2026 at the University of Michigan

due by the end of your lab section on Wednesday, October 28th, 2026

Name: _____

uniqname: _____

Each lab worksheet will contain several activities, some of which will involve writing code and others that will involve writing math on paper. To receive credit for a lab, you must complete all activities and show your lab TA by the end of the lab section.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

Recap: Linear Transformations (Chapter 6.1) and Inverses (Chapter 6.2)

- A linear transformation is a function $\mathbb{R}^d \to \mathbb{R}^n$ such that

$$f(\vec{x} + \vec{y}) = f(\vec{x}) + f(\vec{y}), \qquad f(c\vec{x}) = cf(\vec{x})$$

  - Every linear transformation has a corresponding $n \times d$ matrix A where $f(\vec{x}) = A\vec{x}$.
  - If A is square, then $f(\vec{x}) = A\vec{x}$ is a function from $\mathbb{R}^n$ to $\mathbb{R}^n$, and A is invertible if and only if the function f is invertible.
- An $n \times n$ square matrix A is **invertible** if and only if $\text{rank}(A) = n$, which also means that A's columns are linearly independent (along with several other equivalent conditions).
- If A is invertible, then its **inverse** A^{-1} is the **unique** $n \times n$ matrix such that $AA^{-1} = I = A^{-1}A$.
- The **determinant** of an $n \times n$ square matrix A, denoted $\det(A)$, is defined so that $|\det(A)|$ is the n-dimensional volume of the parallelepiped spanned by the columns of A.
  - If $\det(A) = 0$, then A is not invertible.
  - If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $\det(A) = ad - bc$ and $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.
  - The determinant satisfies several properties, including that $\det(AB) = \det(A)\det(B)$ and $\det(A^T) = \det(A)$.

Activity 1: PrairieLearn Practice Problems

We're testing out a new website for practicing linear algebra problems: PrairieLearn.

On the course website, under Lab 8, click the "PrairieLearn" link to access the relevant problems for this activity. It consists of 6 problems, each worth 1 point. The numbers in the problems are randomized; everyone will receive slightly different problems.

You can also **access the problems here** when reviewing.

To get credit for Activity 1, you must **eventually** correctly answer all 6 problems, earning a score of 6/6. If you answer a problem incorrectly, click "New Variant" to generate a new version and then try again. There is no penalty for answering a problem incorrectly, as long as you eventually get it correct.

To get checked off for Activity 1, show your lab TA that you have earned a score of 6/6 on PrairieLearn.

If you can't access PrairieLearn, post on Ed or email Suraj.

Activity 2: Symbolic Inverses

Given that A is an invertible $n \times n$ matrix that satisfies $A^4 - 3A^2 + 2A - 4I = 0$, find an expression for A^{-1} in terms of A.

Activity 3: Can A Be Invertible?

Suppose A is a 3×3 matrix such that

$$A \begin{bmatrix} 5 \\ -1 \\ 2 \end{bmatrix} = A \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$$

Is A invertible? Why or why not?

Activity 4: Linear Transformations

Suppose $f : \mathbb{R}^3 \to \mathbb{R}^3$ is a linear transformation represented by the matrix A.

Furthermore, suppose that $f\left(\begin{bmatrix}1\\0\\0\end{bmatrix}\right) = \begin{bmatrix}0\\3\\4\end{bmatrix}$, $f\left(\begin{bmatrix}0\\10\\0\end{bmatrix}\right) = \begin{bmatrix}0\\4\\-3\end{bmatrix}$, and $f\left(\begin{bmatrix}0\\0\\1\end{bmatrix}\right) = \begin{bmatrix}1\\0\\0\end{bmatrix}$.

a) Find $f\left(\begin{bmatrix}2\\1\\2\end{bmatrix}\right)$. **After that**, find the matrix A corresponding to f, i.e. where $f(\vec{x}) = A\vec{x}$.

b) Find a **diagonal** matrix D and an **orthogonal** matrix Q such that $A = QD$. (Not every matrix can be written in this form, but this particular A can.) Then, describe **in English** how f transforms a vector $\vec{x}$.

c) Using your $A = QD$ decomposition from part **b)**, find A^{-1}.

Hint: Recall that for orthogonal matrices, $QQ^T = Q^TQ = I$. And, for any invertible matrices A and B, $(AB)^{-1} = B^{-1}A^{-1}$.

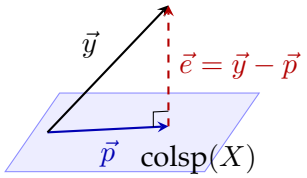
d) Using the scaling factors from part **b)**, find $|\det(A)|$. (You can verify your work using the formula in **Chapter 6.1**.)

Activity 5: Projecting onto the Column Space

The next big idea in the course is finding the projection of a vector $\vec{y} \in \mathbb{R}^n$ onto the column space of an $n \times d$ matrix X.

Let

$$X = \begin{bmatrix} 2 & 1 \\ 0 & -3 \\ 0 & 0 \end{bmatrix}, \qquad \vec{y} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$



a) Fill in the blank: colsp(X) is a _____ in $\mathbb{R}^3$.

b) Which vector in colsp(X) is closest to $\vec{y}$? Normally, this would require more calculation, but it is possible to see the answer here without much work. Explain your thought process.

We'll call the vector you found $\vec{p}$.

c) Find the error vector $\vec{e} = \vec{y} - \vec{p}$.

d) Verify that $\vec{e}$ is orthogonal to each column of X.

e) Find $\vec{w}^* \in \mathbb{R}^2$ such that $X\vec{w}^* = \vec{p}$.

We were able to find $\vec{p}$ without much calculation because of the particular X and $\vec{y}$ we chose. But we need a **more general-purpose solution**, one that works for any $n \times d$ matrix X and any $\vec{y} \in \mathbb{R}^n$. We can't rely on being able to see the answer, especially when $n > 3$!

Important: Every vector in colsp(X) can be written as $X\vec{w}$ for some $\vec{w} \in \mathbb{R}^d$.

Our goal is to find $\vec{w}$ so that the error $\vec{e} = \vec{y} - X\vec{w}$ is orthogonal to every column of X. Let $\vec{x}^{(1)}, \ldots, \vec{x}^{(d)}$ be the columns of X.

f) Fill in the blank: For $\vec{e}$ to be orthogonal to every column of X, we need

$$\vec{x}^{(1)} \cdot \vec{e} = 0, \quad \ldots, \quad \vec{x}^{(d)} \cdot \vec{e} = 0,$$

which we can write as a single matrix-vector product:

$$_____ \; \vec{e} = \vec{0}$$

g) Substitute $\vec{e} = \vec{y} - X\vec{w}$ into your equation and rearrange it to fill in the blanks below. This is called the **normal equation**.

$$\underbrace{\qquad\qquad}_{d \times d} \vec{w} = \underbrace{\qquad\qquad}_{d \times 1}$$

The normal equation describes the condition we want for any $n \times d$ matrix X. To solve it using an inverse, we'll now assume that X's columns are linearly independent.

h) Explain why $X^T X$ is invertible under this assumption. *Hint: Recall that rank$(X^T X) =$ rank(X), as explained in* ***this video****.*

i) Solve the normal equation for $\vec{w}$ by multiplying both sides by an appropriate inverse. Call your solution $\vec{w}^*$.

The closest vector in colsp(X) to $\vec{y}$ is $\vec{p} = X\vec{w}^*$. We call $\vec{p}$ the **projection** of $\vec{y}$ onto colsp(X).

Now, let's try our method with a different X:

$$X = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 0 \end{bmatrix}, \qquad \vec{y} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

j) Use the normal equation to find $\vec{w}^*$.

k) Find the projection $\vec{p}$ of $\vec{y}$ onto colsp(X).

l) Find the error vector $\vec{e} = \vec{y} - \vec{p}$.

m) Why do the components of $\vec{e}$ sum to 0 in this example, but not in the first example in this activity?